

CONTROLLED PILOT / INTERMEDIATE WORKFLOW

Vector conversion and batch release workflow

One exemplar has been converted from the approved raster into SVG, EPS, and PDF masters. The future queue is 100-500 items. This workflow defines when automatic tracing may pass, when manual cleanup is mandatory, and when production must stop for approval.

EXEMPLAR APPROVAL: PENDING

BATCH RELEASE: BLOCKED

Pilot decision

The whole-frame automatic trace was rejected for final use. The interwoven multicolor emblem, distinctive display lettering, and raster paper/ink noise would create false islands and excess nodes. The candidate master instead uses measured geometric shapes, clean Bezier emblem paths, a cleaned source-derived wordmark outline, and outlined supporting glyphs.

1. Ingest + hash	2. Auto candidate	3. Measure	4. Accept or clean	5. Export + render	6. Human gate
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Release record required

The exemplar cannot release the batch until the record contains approver name, approval timestamp, approved master SHA-256, and approved preview SHA-256. Any later master change invalidates that approval.

Automatic tracing acceptance

Automatic tracing is accepted only when every hard gate and every numeric threshold passes. A single failure routes the item to manual review; averages cannot hide a local defect.

Gate / metric	Acceptance requirement
Wording	OCR and operator check exactly match all approved wording, punctuation, spacing, and weight labels.
Raster dependency	Zero embedded raster images in SVG, EPS, and PDF masters.
Geometry	Silhouette IoU at least 0.985; 95th-percentile edge distance no more than 0.30 mm.
Color	Median Delta E 2000 no more than 2.0; 95th percentile no more than 4.0; every color mapped to an approved palette role.
Node economy	Logo no more than 3.0 nodes/mm; display outlines no more than 8.0 nodes/mm after cleanup.
Path integrity	Zero unintended open fills, self-intersections, lost counters, duplicate paths, or unintended path-count changes.
Small detail	All intended strokes and gaps remain at least 0.20 mm at nominal output size.
Noise	Zero micro-islands smaller than 0.05 mm ² after cleanup.

Required scale renders

Render every item at small-use size, nominal print size, and a large inspection scale. Check counters, crossings, strokes, rule alignment, and text legibility at each scale before export hashes are recorded.

Manual review triggers

These conditions require operator review and a recorded trigger code. The operator may clean nodes, rebuild geometry, or request a better source; the approved identity may not be restyled.

Code	Trigger
OCR_MISMATCH	Any wording, punctuation, spacing, or weight label differs.
INTERWEAVE_OR_OCCLUSION	Over-under crossings, overlaps, masks, or occluded shapes are present.
UNKNOWN_DISPLAY_TYPE	A distinctive display face cannot be matched to an approved font file.
TRACE_COMPLEXITY	Node density exceeds the relevant automatic threshold.
PALETTE_DRIFT	Median or p95 Delta E exceeds the automatic threshold.
GEOMETRY_DRIFT	Silhouette IoU or edge-distance threshold fails.
SMALL_DETAIL	A feature or stroke is below 0.20 mm at nominal size.
RASTER_NOISE	Texture, compression, antialiasing, or gradient noise creates false islands.
PATH_DEFECT	Lost counters, self-intersections, unintended open paths, or duplicate paths occur.
LOW_SOURCE_RESOLUTION	Effective source resolution is below 150 ppi at nominal size.

Per-item audit record

item_id, source_sha256, operator, trace_engine_version, threshold_results, manual_review_codes, node_counts_before_after, palette_values, export_sha256s, preview_paths, approval_status, approver, approval_timestamp.

Stop condition

If the source identity cannot be recovered without guessing, record the blocker and stop. Do not invent missing geometry, wording, palette values, or client approval.